

Zero and Negative Exponents

Evaluating zero and negative powers, rewriting expressions with positive exponents, applying the exponent laws, and comparing powers

Directions.

- Two definitions do all the work: $a^0 = 1$ for every $a \neq 0$, and $a^{-n} = \frac{1}{a^n}$.
- A negative exponent never makes a value negative. It means *reciprocal*.
- Give numeric answers as exact fractions in lowest terms, and give algebraic answers with **positive exponents only**.
- Show the law you used (product, quotient, or power of a power) before you simplify.

1. Evaluate. 7^0

Answer: _____

2. Evaluate. Write your answer as a fraction. 5^{-2}

Answer: _____

3. Evaluate. Write your answer as a fraction. $\left(\frac{2}{3}\right)^{-2}$

Answer: _____

4. Rewrite with positive exponents only. $3x^{-2}$

Answer: _____

5. Simplify using the product law, and write the result with a positive exponent. $x^{-8} \cdot x^5$

Answer: _____

$$2^{-3} \quad \underline{\hspace{1cm}} \quad 3^{-2}$$

Answer: _____

Answer: _____

8. Simplify, and write the result with a positive exponent. $(x^{-2})^3 \cdot x^4$

Answer: _____

9. Rewrite with positive exponents only, then simplify.

$$\frac{2x^{-3}}{y^{-2}}$$

Answer: _____

10. Order these three powers from **least to greatest**. Write the value of each one first.

$$5^0, \quad 5^{-1}, \quad 5^1$$

Answer: _____

11. Find and fix the mistake. Dana was asked to evaluate 4^{-2} and wrote -16 . Explain in one sentence what Dana did wrong, then write the correct value as a fraction.

Answer: _____

12. Simplify completely. Your answer must have positive exponents only.

$$\frac{3x^{-2}y^4}{6x^3y^{-1}}$$

Answer: _____